



Executive M.tech - Data Science and Artificial Intelligence

Batch – 2: Course Name - Python for Data Science

Project Report

#NOTE: Save the file with your SRN_NAME and convert to PDF and Submit

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Module Assigned: python-docx

Module Details:

Library Name	python-docx
Version	0.8.11
Official Website	python-docx — python-docx 1.2.0 documentation

Purpose of this Module (One paragraph):

python-docx is a Python library that allows developers to create, read, edit, and save Microsoft Word (.docx) documents programmatically. It is commonly used to automate document-related tasks such as generating reports, invoices, certificates, resumes, and other formatted documents without manually using Microsoft Word. The library provides features for adding text, headings, tables, images, page breaks, and formatting, making it a useful tool for document automation and data-driven report generation in Python applications.

Features implemented by the Learner (Minimum 10):

1. Create a new .docx document and initialize document state
2. Add headings (multiple levels: H1, H2, H3)
3. Add and edit paragraphs
4. Apply character and paragraph formatting (bold, italic, underline, size, color, spacing)
5. Adjust font options (typeface, size, color)
6. Set paragraph alignment (left, center, right, justify)
7. Add unordered (bulleted) lists
8. Add ordered (numbered) lists
9. Insert and populate tables with data (up to 10 × 10, CSV-style input)
10. Save and export the .docx file (one-click download + Clear All)

11. Insert images (PNG, JPG, GIF, BMP) with configurable width

Industry Applications (Minimum 5):

1. **Automated Report Generation** — finance and analytics teams generate recurring reports (P&L, KPIs, balance sheets) directly from databases without manual Word editing; tools like Power BI and Tableau export pipelines use similar techniques
2. **HR Documents** — automatically produce offer letters, contracts, salary slips, and onboarding packs populated from HRMS systems (SAP SuccessFactors, Workday)
3. **Legal** — assemble templated contracts, NDAs, compliance documents, and court filings at scale; DocuSign and Clio use document generation under the hood
4. **Education** — generate personalised assignments, certificates, progress reports, and feedback forms for large student cohorts from LMS data (Moodle, Canvas)
5. **Healthcare** — produce patient discharge summaries, lab reports, and prescription documents auto-filled from EHR systems (Epic, Cerner)

Tool / Platform	How document generation is used
Django / Flask web apps	Generate downloadable Word reports on demand for end users
Jupyter Notebooks	Export analysis results and charts as formatted .docx reports
Airflow / Prefect pipelines	Schedule and auto-distribute Word reports as pipeline outputs
Pandas + python-docx	Convert DataFrame analysis results into formatted Word tables
ReportLab / docxtpl	Companion libraries for more complex templated Word generation
Zapier / Make (Integromat)	Trigger Word document creation from form submissions or CRM events
Microsoft Power Automate	Similar automated document generation workflows in enterprise environments

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